03_nurse_rostering_case_study

Nurse Rostering & Acuity-to-Staffing

Synthetic Data Case Study

Synthetic Australian Hospital Operations Dataset Bundle

Executive Summary

This case study walks through a sample analysis of the Nurse Rostering & Acuity-to-Staffing dataset — 32,850 synthetic shift-level roster records across 8 fictional facilities and 10 ward/unit types. It demonstrates the kind of staffing-gap, acuity, and overtime-modelling analysis this dataset supports. All figures below are calculated directly from the dataset.

⚠ **This is a synthetic dataset. Every figure below describes a simulated pattern, not a real-world workforce-planning finding.**

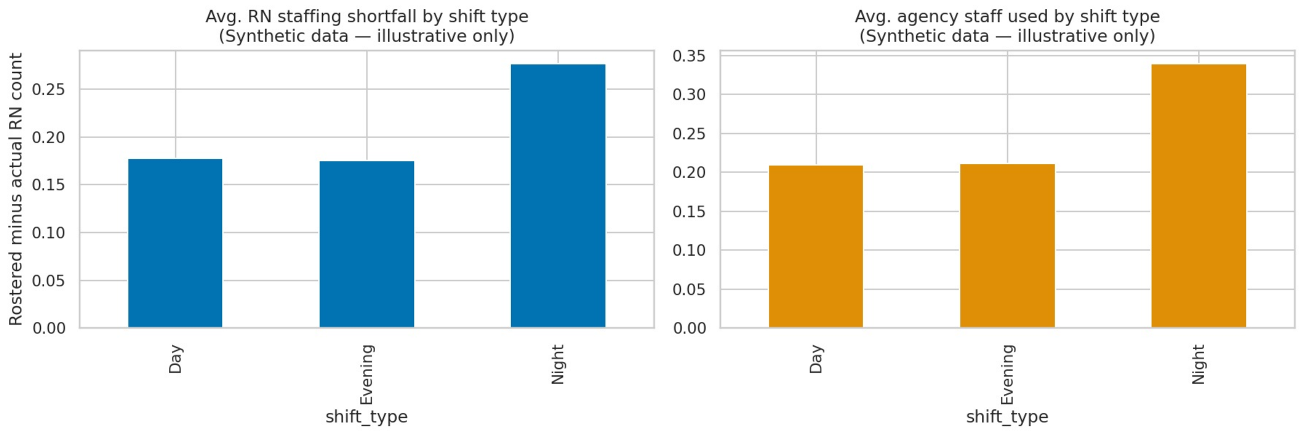
Dataset Snapshot

- **32,850** shift-level roster records, **19** columns
- **8** fictional facilities, **10** ward/unit types
- Understaffed-shift rate: **23.35%** overall
- Date range: 2025-01-01 to 2025-12-31

Key Findings (in this synthetic model)

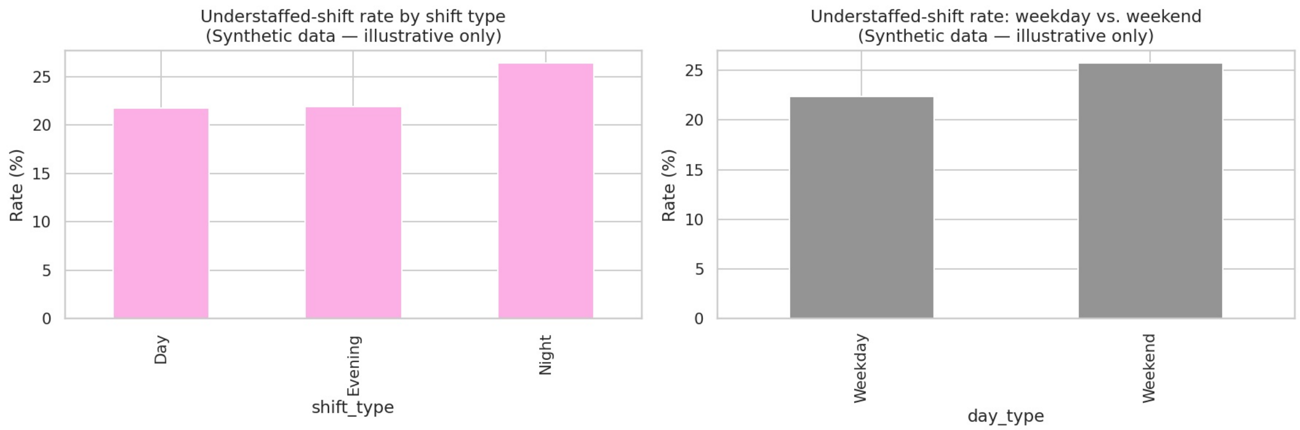
- **Night shift shows the highest modelled understaffed rate** — 26.41%, compared with 21.73% on day shift and 21.92% on evening shift.
- **Overtime hours roughly double on understaffed shifts** — averaging 2.67 hours, versus 1.33 hours on adequately staffed shifts, reflecting the modelled backfill chain (sick leave → agency use → overtime).
- **Recommended nurse:patient ratios vary sharply by ward type by design** — intensive-care and coronary-care units are modelled with far more intensive staffing ratios than rehabilitation or aged-care wards.
- The **understaffed flag is fully auditable**: it is derived directly from actual vs. recommended ratio (flagged when actual exceeds recommended by more than 15%), not an independently-randomised field.

Visual Highlights



Staffing gap

Average RN staffing shortfall and agency staff use by shift type — night shift shows the largest modelled gap.



Shift patterns

Understaffed-shift rate by shift type and by weekday vs. weekend.

Methodology Note

Actual staffing is modelled as rostered staffing minus a sick-leave effect (higher on nights and weekends), partially backfilled by agency staff and overtime. These are designed relationships built into the data generator — not measurements of any real health service's rostering data. Full methodology and limitations: see `methodology_bias_limitations.md`.

Why This Dataset Is Useful

This dataset gives analysts, students, and QA/software testers a realistic-feeling, richly- structured workforce/rostering dataset to practise on without any real staff privacy risk. It supports:

- SQL and Python staffing-gap and overtime-analysis exercises
- Power BI/Tableau dashboard-building practice (understaffing by ward, shift, day type)
- An educational classification exercise (understaffing prediction) with genuine class imbalance
- Portfolio-ready case studies and dashboards — like this one

Restriction / Disclaimer

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